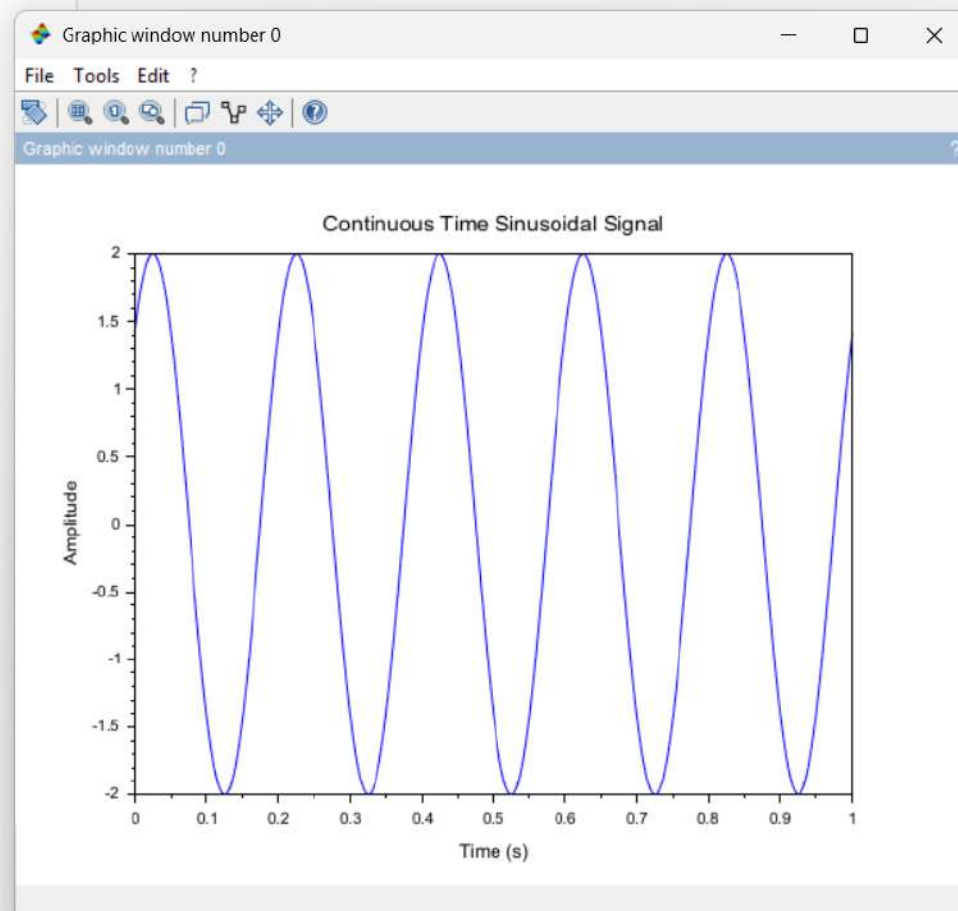


```
1 clc;
2 clear;
3 clf;
4
5 //Define parameters
6 f = 5; //Frequency in Hz
7 A = 2; //Amplitude
8 phi = -pi/4; //Phase shift in radians
9
10 //Time vector
11 t = 0:0.001:1;
12
13 //Generate sinusoidal signal
14 x_t = A * sin(2 * pi * f * t + phi);
15
16 //Plot the signal
17 plot(t, x_t);
18 xlabel("Time (s)");
19 ylabel("Amplitude");
20 title("Sinusoidal Signal");
21 xtitle("Continuous Time Sinusoidal Signal", "Time (s)", "Amplitude");
22
```



EXP_12 SS.sci (E:\EXP_12 SS.sci) - SciNotes

EXP_12 SS.sd

```
1 //Test Case: Discrete-Time Unit Impulse Signal Generation.
2 //Define parameters.
3 N = 21; ..... // Length of the signal
4 n0 = 11; ..... // Index at which the impulse occurs
5 n = 1:N; ..... // Discrete time index vector.
6 //Generate the unit impulse signal.
7 x_n = zeros(1, N); ..... // Initialize signal with zeros.
8 x_n(n0) = 1; ..... // Set the impulse at the desired location.
9 //Plot the signal
10 clf; //Clear the current figure.
11 plot2d(n, x_n);
12 xlabel('n - (Discrete-Time Index)');
13 ylabel('Amplitude');
14 title('Discrete-Time Unit Impulse Signal');
15 xtitle('Discrete-Time Unit Impulse Signal', 'n - (Discrete-Time Index)', 'Amplitude');
16
```

